lying on a red tendex line. It stretches her, too; she feels a stretching force between her head and her feet, exerted by the red tendex line.

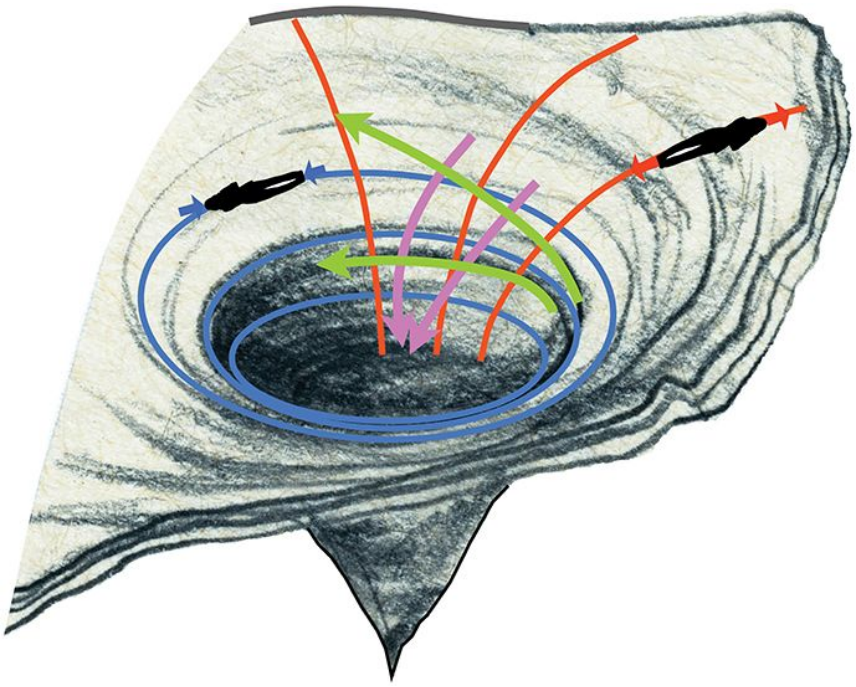


Fig. 4.7. Tendex lines around a black hole. The picture of the hole is extracted from Lia Halloran's drawing, Figure 4.5.

The purple paths begin, at their top ends, running parallel to each other. They are then squeezed together by the blue tendex lines, and the woman whose body lies along a blue tendex line is also squeezed.

This stretching and squeezing is just a different way of thinking about the influence of the warping of space and time. From one viewpoint, the paths are stretched apart or squeezed together due to the planetary paths moving along the straightest routes possible in the warped space and time. From another viewpoint it is the tendex lines that do the stretching and squeezing. Therefore, the tendex lines must, in some very deep way, represent the warping of space and time. And indeed they do, as the mathematics of the Riemann tensor taught us.

Black holes are not the only objects that produce stretching and squeezing forces. Stars and planets and moons also produce them. In